

3ART13 LED

Variant: CHECKED

2026-05-21

Rev 2.1.0+ (Unreleased)

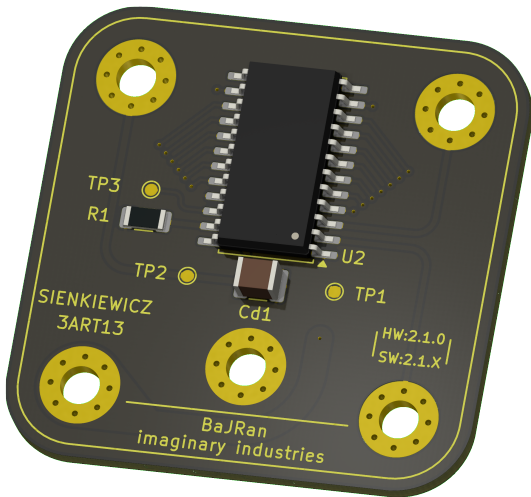
Page	Index	Page	Index
1	Cover Page	11	
2	Block Diagram	12	
3	Project Architecture	13	
4	Sec A - Shift Register Screw Interface	14	
5	Sec B - Shift Register	15	
6	Sec C - LED Display	16	
7	Power - Sequencing	17	
8	Revision History	18	
9		19	
10		20	

Page	Index
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 21-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

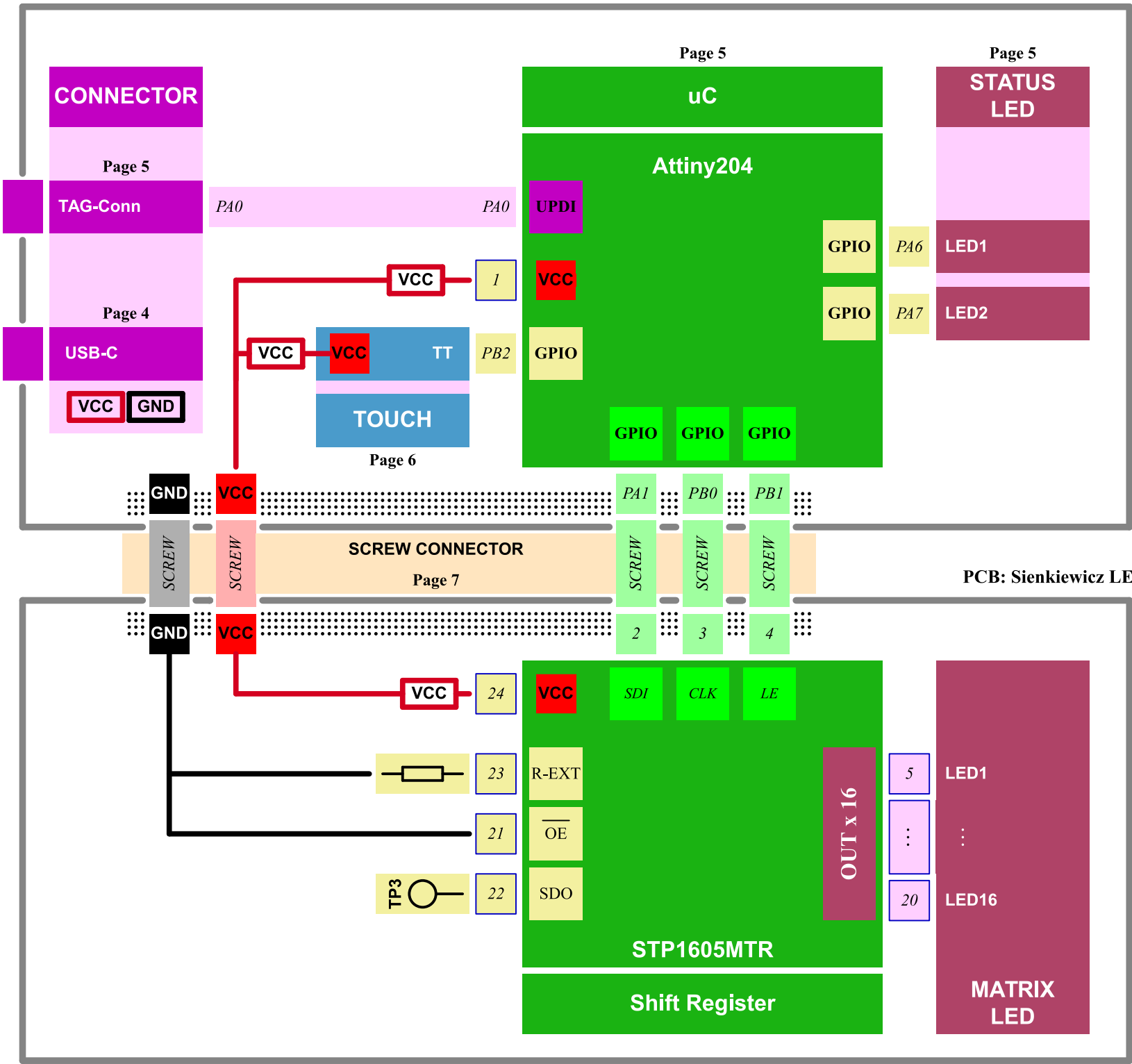
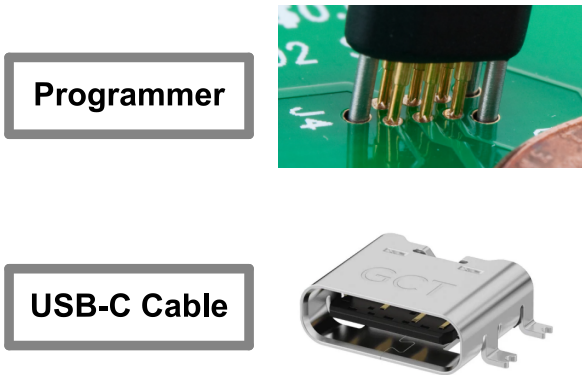
LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: BaJRan imaginary industries	Variant: CHECKED	Git Hash: ff91115
	Sheet Title:	Board Name: 3ART13 LED	Project Name: Sienkiewicz	
	Sheet Path: /	File Name: Sienkiewicz_LED.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
			Reviewer:	Revision: 2.1.0+ (Unreleased)
			Size: A3	Sheet: 1 of 8

[2] Block Diagram

PCB: Sienkiewicz TOUCH

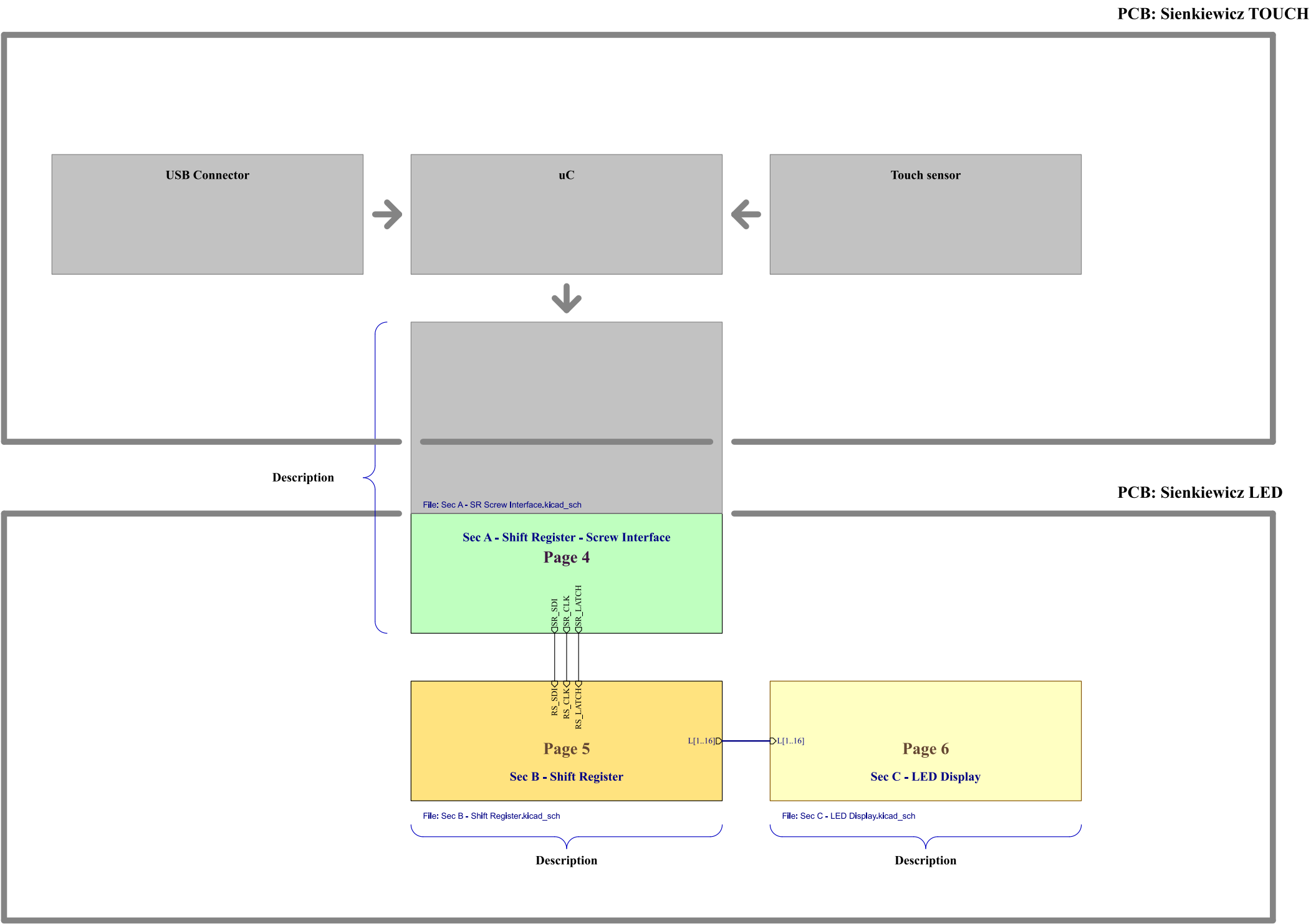


Target specifications:

Input voltage	5V
Current	200mA
Spec 3	----
Spec 4	----

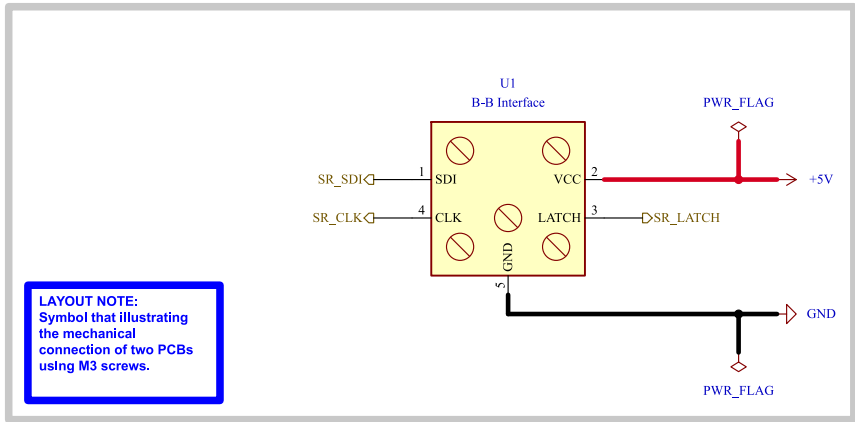
	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
	Sheet Title: Block Diagram	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Path: /Block Diagram/	File Name: Block Diagram.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 2 of 8

[3] Project Architecture



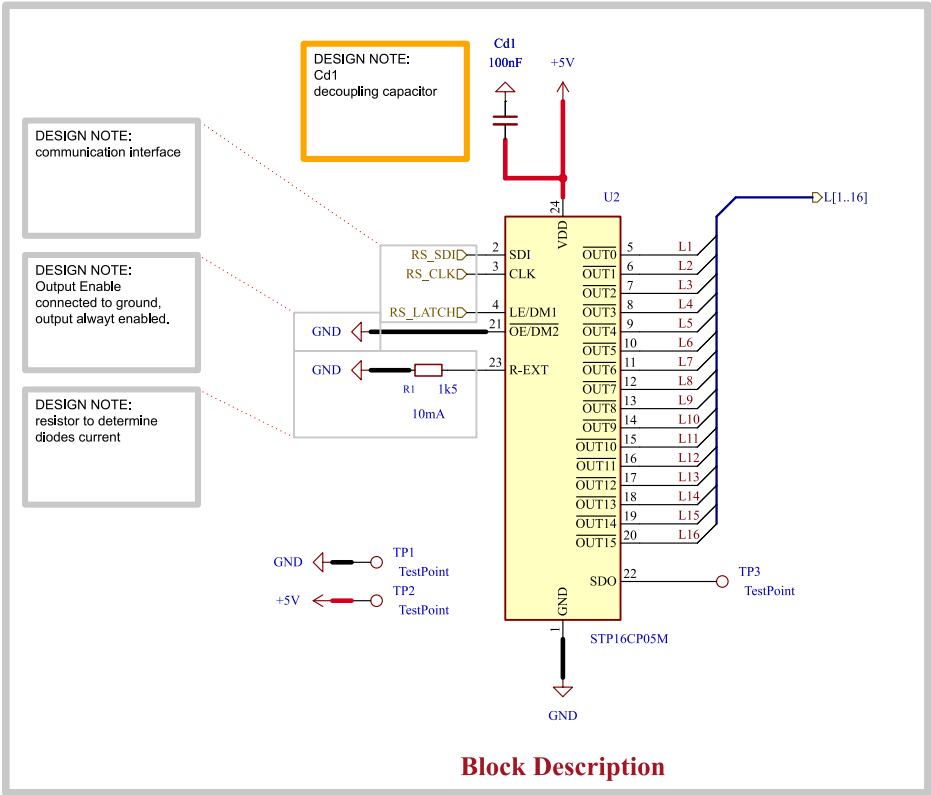
	Comments:	Company: BaJRan  imaginary industries		Variant: CHECKED	Git Hash: ff91115	
		Board Name: 3ART13 LED		Project Name: Sienkiewicz		
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)	
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 8	

[4] Shift register - Screw Interface



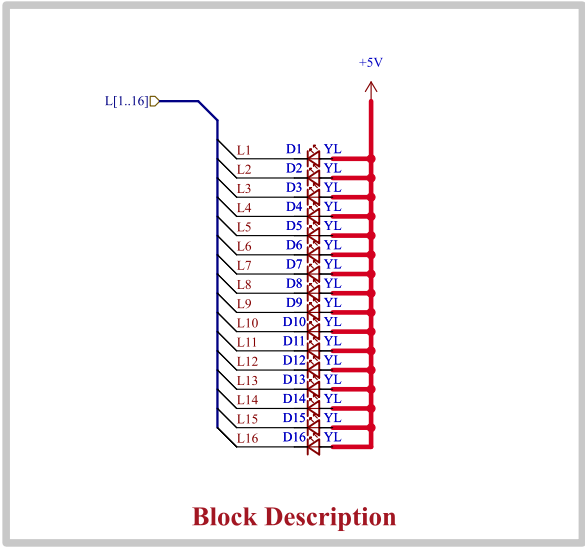
	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
	Sheet Title: Shift register - Screw Interface	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Path: /Project Architecture/Sec A - Shift Register - Screw Interface/	File Name: Sec A - SR Screw Interface.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
			Reviewer:	Size: A4	Sheet: 4 of 8

[5] Shift Register



	Comments:		Company:		Variant:		Git Hash:			
			BaJRan 		CHECKED		f911115			
			Board Name:			Project Name:				
			3ART13 LED			Sienkiewicz				
	Sheet Title:		File Name:		Designer:		Date:		Revision:	
Shift Register		Sec B - Shift Register.kicad_sch		Pawel Baran		2025-01-12		2.1.0+ (Unreleased)		
Sheet Path:		/Project Architecture/Sec B - Shift Register/			Reviewer:		Size:		Sheet:	
							A4		5 of 8	

[6] LED Display



Block Description

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: LED Display	File Name: Sec C - LED Display.kicad_sch	Designer: Pawel Baran	Date: 2026-04-30	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Project Architecture/Sec C - LED Display/		Reviewer:	Size: A4	Sheet: 6 of 8

[7] Requirements

Project	Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering.	MEET	The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector.
MAIN OBJECTIVE			

1	The project shall contain very small integrated circuits. Pitch less than 1mm	MEET	IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95 IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27
2	The project shall contain 0805 SMT passives components.	MEET	Device contain multiple 0805 SMT elements.
3	The project shall contain multi-pin integrated circuits to verify soldering precision.	MEET	IC: Shift register : STP16CP05MTR : SO-24 : 24 pins IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins
4	The project shall contain an UPDI programming interface for firmware upload and debugging.	MEET	uC Attiny204 with UPDI programming interface
5	Possibility to program using Microchip enviroment and pickit-4 programmer.	MEET	It is possible to program the device using pickit-4 and MPLAB software.
6	Shall contain at least two status led.	MEET	Status LED'S : LED1, LED2.
7	Shall be able to interact with user.	MEET	The user can change the light effect using a capacitive touch sensor.
8	Shall be easy to power up.	MEET	Powered by USB-C connector with general mode
9			
10			
11			
12			

		Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
			Board Name: 3ART13 LED		Project Name: Sienkiewicz	
		Sheet Title: Requirements	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
		Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 7 of 8

[8] Revision History

Version 2.1.0 - 2026-05-02

- Added
- First version of PCB and Schematic

Version 2.1.1 not found.
Version 2.1.1 not found.

	Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
		Board Name: 3ART13 LED		Project Name: Sienkiewicz	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8